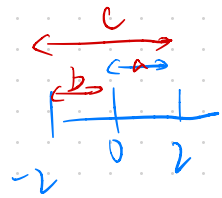
1–4 A particle moves according to a law of motion $s = f(t)$, $t \geqslant 0$, where t is measured in seconds and s in feet.

(a) Find the velocity at time t.
(b) What is the velocity after 1 second?
(c) When is the particle at rest?
(d) When is the particle moving in the positive direction?
(e) Find the total distance traveled during the first 6 seconds.
(f) Draw a diagram like Figure 2 to illustrate the motion of the particle.
(g) Find the acceleration at time t and after 1 second.
(h) Graph the position, velocity, and acceleration functions for $0 \leqslant t \leqslant 6$.

$s = f(t) = t^3 - 9t^2 + 24t$



1. a) $v = f'(t) = 3t^2 - 18t + 24$

b) $v(1) = 3 - 18 + 24 = 9$

c) $v(t) = 0$

$3t^2 - 18t + 24 = 0$

$(3t - 6)(t - 4) = 0$

$t = 2$ or $t = 4$

d) $t^3 - 9t^2 - 24t > 0$ $v > 0$ 往正方向

$3t^2 - 18t + 24 > 0$

$t \in (0, 2) \cup (4, +\infty)$

e) $s = 2^3 - 9 \times 2^2 + 24 \times 2 - (4^3 - 9 \times 4^2 + 24 \times 4) + 6$ $c = a - b$

$s = |f(2) - f(0)| + |f(4) - f(2)| + |f(6) - f(4)|$

相对上次移动了多少

(i) When is the particle speeding up? When is it slowing down?

1. $f(t) = t^3 - 9t^2 + 24t$ **2.** $f(t) = \dfrac{9t}{t^2 + 9}$

3. $f(t) = \sin(\pi t/2)$ **4.** $f(t) = t^2 e^{-t}$

1. $v(t) = f'(t) = 3t^2 - 18t + 24$

$a(t) = v'(t) = 6t - 18 \quad > 0$

$t > 3$

$3t = 6$

$t = 2$

$v(t) = 3t^2 - 18t + 24 > 0$

$(3t - 6)(t - 4) > 0$

	$t<2$	$2<t<4$	$t>4$
$t-4$	$-$	$-$	$+$
$3t-6$	$-$	$+$	$+$

when $0 < t < 2$ or $t > 4$, $v > 0$, moving to the right

$2 < t < 4$, moving to the left

same sign. S.D.

	$0<t<2$	$2<t<3$	$3<t<4$	$t>4$
$v(t)$	$-$	$-$	$+$	$+$
$a(t)$	$+$	$-$	$-$	$+$
	S.D	S.U	S.D	S.U

when $2 < t < 3$ or $t > 4$ the particle speeding up

when $0 < t < 2$ or $3 < t < 4$ slowing down